

Post-Workout Recovery Templates

Mealvana Endurance

What the Experts Say

Dr. Rachel Mitchell, PhD (Nutrition Science)

Dr. Mitchell is a nutrition science advisor for Mealvana. Her rule for post-workout recovery:

“Any combo works well as long as both a carb source and protein source is covered. The 3-4:1 ratio of carbs to protein is ideal.”

Her recommended post-workout formulas:

Carb Source	Protein Source
Chocolate milk	(built-in)
Granola + honey + berries	Greek yogurt
Banana + peanut butter + graham crackers	Protein shake
Any fruit	Protein shake
Granola bar	Protein shake
Toast + jam	Eggs
Applesauce	Cottage cheese
Applesauce + graham crackers	Protein shake
PB&J sandwich	Glass of milk
Crackers + jam	Cheese stick
Oatmeal + honey (or maple syrup) + banana	Protein powder + almond butter
Fruit smoothie	Greek yogurt + protein powder
Bagel + cream cheese	Turkey slices
Waffle + banana	Protein shake
Toast + honey	Protein shake
Crackers	Turkey + cheese roll-ups

What Pro Athletes Eat After Training

Jan Frodeno (3x Ironman World Champion): Banana + protein powder + milk + peanut butter blended. Same smoothie every single day.

Lucy Charles-Barclay (Ironman World Champion): Protein shake within 30 minutes of every session – a team member has it waiting at the finish line for races.

Lionel Sanders (Pro Triathlete): Core Power protein shakes for travel convenience. Smoothie bowls at home. Reformed sugar intake after bloodwork showed pre-diabetic levels despite world-class fitness.

Sam Long (Pro Triathlete): Eggs. “The single food item he could not live without.” Same recovery foods daily to eliminate decision fatigue.

Cody Beals (Pro Vegan Triathlete): Vega Sport plant protein shake immediately after every session.

The 10 Recovery Templates

Every template includes a carb source and a protein source. Hydration (water) and electrolytes are added separately by the system based on the athlete's sweat rate and conditions.

Template 1: Chocolate Milk + Banana + Protein Powder

The most studied recovery combination. Protein powder mixed into the milk closes the protein gap for larger athletes or longer sessions.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Chocolate Milk	1 cup (8 oz)	30 g	9 g	160 mg	3 cups
Banana	1 medium	27 g	1.3 g	1 mg	2
Protein Powder	1 scoop	3 g	24 g	130 mg	2 scoops

Scaling: Chocolate milk and banana scale together to hit carb targets. Protein powder is fixed at 1 scoop for most athletes (2 scoops only for 90+ kg athletes after long sessions).

Allergens: dairy | **Excludes:** vegan, paleo

Template 2: Protein Shake + Banana + Graham Crackers

Grab-and-go from any convenience store. No blender, no cooking. Graham crackers deliver satisfying crunch after hours of gels and liquids.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Protein Shake (RTD bottle)	1 bottle (11 oz)	5 g	25 g	200 mg	1 bottle
Banana	1 medium	27 g	1.3 g	1 mg	2
Graham Crackers	2 sheets	21.5 g	2 g	169 mg	2 servings

Scaling: Protein shake is always 1 bottle (fixed). Banana and graham crackers scale up for higher carb targets. A 91 kg athlete after a long ride might get 2 bananas + 4 sheets of graham crackers alongside the same 1 bottle.

Allergens: dairy, gluten | **Excludes:** gluten-free

Template 3: Recovery Protein Drink + Pretzels

Zero prep. Recovery protein drink (like Core Power) from a gas station cooler plus pretzels. Lionel Sanders' travel go-to.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Recovery Protein Drink	1 bottle (14 oz)	9 g	42 g	250 mg	3 bottles
Pretzels	1 oz	23 g	2.8 g	385 mg	4 oz

Scaling: These scale independently. The drink is typically 1 bottle (protein is already met). Pretzels scale freely to hit carb targets – a bigger athlete or longer workout just means more pretzels. For very long sessions (3+ hr), a second bottle can be added.

Allergens: dairy, gluten | **Excludes:** vegan, paleo, gluten-free

Template 4: Plant Protein Shake + Banana + Pretzels

Fully vegan, grab-and-go. Cody Beals' approach – plant protein shake immediately after every session.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Plant Protein Shake (RTD bottle)	1 bottle (11 oz)	10 g	20 g	320 mg	1 bottle
Banana	1 medium	27 g	1.3 g	1 mg	2
Pretzels	1 oz	23 g	2.8 g	385 mg	4 oz

Scaling: Shake is fixed at 1 bottle. Banana and pretzels scale independently to hit carb targets. After a short easy workout, might just be 1 banana + no pretzels. After a long hard session, 2 bananas + 2-3 oz pretzels.

Allergens: gluten | **Excludes:** gluten-free | **Vegan-friendly**

Template 5: Yogurt + Granola + Fruit + Honey + Protein Powder

The parfait. Assemble in a mason jar the night before and grab from the fridge. Protein powder stirred into yogurt ensures the protein target is met.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Yogurt (plain)	1 cup	17.4 g	8.5 g	172 mg	2 cups
Granola	1/4 cup	17.5 g	3 g	7 mg	2 servings
Banana	1 medium	27 g	1.3 g	1 mg	2
Honey	1 tbsp	17.3 g	0.1 g	1 mg	2 tbsp
Protein Powder	1 scoop	3 g	24 g	130 mg	2 scoops

Scaling: Yogurt, granola, banana, and honey scale together proportionally (bigger parfait). Protein powder is fixed at 1 scoop. Swap banana for mixed berries, blueberries, or any seasonal fruit.

Allergens: dairy, gluten | **Excludes:** vegan, paleo

Template 6: Oatmeal + Protein Powder + Banana + Honey

Warm and comforting. Add hot water, stir in protein powder, top with banana and honey. Dr. Mitchell recommends multiple oatmeal variations.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Oatmeal (1/2 cup dry)	1 serving	27 g	5 g	0 mg	2 servings
Protein Powder	1 scoop	3 g	24 g	130 mg	2 scoops
Banana	1 medium	27 g	1.3 g	1 mg	2
Honey	1 tbsp	17.3 g	0.1 g	1 mg	2 tbsp

Scaling: Oatmeal, banana, and honey scale together (bigger bowl). Protein powder is fixed at 1 scoop. Swap honey for maple syrup. Add almond butter or peanut butter for extra calories and healthy fats.

Allergens: dairy, gluten | **Excludes:** vegan (use plant protein powder for vegan version), gluten-free

Template 7: PB&J + Milk

Under \$2 total. Allen Lim (Skratch Labs) calls PB&J “the perfect real food recovery.” Dr. Mitchell lists both “PB&J sandwich” and “PB&J + glass of milk.”

Food	Serving	Carbs	Protein	Sodium	Max Servings
Toast / Bread	1 slice	12.7 g	1.8 g	130 mg	2 slices
Peanut Butter	1 tbsp	3.6 g	3.6 g	73 mg	2 tbsp
Jam / Jelly	1 tbsp	13.8 g	0.1 g	6 mg	2 tbsp
Milk (whole)	1 cup (8 oz)	12 g	8 g	105 mg	3 cups

Scaling: All components scale together as a unit. A bigger athlete gets a bigger sandwich (2 slices, 2 tbsp PB, 2 tbsp jam) and more milk. Think of it as “number of PB&J sandwiches + glasses of milk.”

Allergens: gluten, peanuts, dairy | **Excludes:** vegan, nut-free

Template 8: Bagel + Cream Cheese + Turkey

Savory recovery for athletes who crave salt after hard sessions. Dense carbs, lean protein, and lots of sodium.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Bagel (large)	1 large	53 g	11 g	462 mg	2
Cream Cheese	1 tbsp	0.8 g	0.9 g	47 mg	2 tbsp
Turkey (deli)	3 oz	2 g	15 g	744 mg	2 servings

Scaling: All components scale together. One bagel = one serving of cream cheese + one serving of turkey. A 91 kg athlete after a long workout might get 1.5 bagels (or 1 large + half) with proportionally more cream cheese and turkey.

Allergens: gluten, dairy | **Excludes:** vegan, vegetarian, gluten-free

Template 9: Toast + Egg + Jam

The morning workout recovery breakfast. Sam Long's eggs are his #1 recovery food. Quick to prepare – boil eggs in advance or scramble in 3 minutes.

Food	Serving	Carbs	Protein	Sodium	Max Servings
Toast / Bread	1 slice	12.7 g	1.8 g	130 mg	2 slices
Egg (boiled/poached)	1 large	1 g	6.1 g	62 mg	2 eggs
Jam / Jelly	1 tbsp	13.8 g	0.1 g	6 mg	2 tbsp

Scaling: All components scale together. More toast = more eggs = more jam. Two slices of toast with 2 eggs and 2 tbsp jam is the standard “full serving.”

Allergens: gluten, eggs | **Excludes:** vegan | **Dairy-free**

Template 10: Waffle + Banana + Protein Shake

Fun and fast. Pop waffles in the toaster, grab a shake from the fridge, eat a banana. Done in 2 minutes. Dr. Mitchell’s “waffle + protein shake + banana.”

Food	Serving	Carbs	Protein	Sodium	Max Servings
Toaster Waffle	1 waffle	15.4 g	2 g	212 mg	2 waffles
Banana	1 medium	27 g	1.3 g	1 mg	2
Protein Shake (RTD bottle)	1 bottle (11 oz)	5 g	25 g	200 mg	1 bottle

Scaling: Waffles and banana scale together to hit carb targets. Protein shake is always 1 bottle (fixed). A bigger target means more waffles and/or an extra banana alongside the same single bottle.

Allergens: gluten, eggs, dairy | **Excludes:** vegan, gluten-free

Dietary Compatibility

#	Template	Vegan	Dairy-Free	Gluten-Free	Nut-Free	Egg-Free
1	Choc Milk + Banana + Protein Powder			Yes	Yes	Yes
2	Protein Shake + Banana + Grahams				Yes	Yes
3	Recovery Drink + Pretzels				Yes	Yes
4	Plant Protein + Banana + Pretzels	Yes	Yes		Yes	Yes
5	Yogurt Parfait + Protein Powder				Yes	Yes
6	Oatmeal + Protein + Banana + Honey				Yes	Yes
7	PB&J + Milk					Yes
8	Bagel + Cream Cheese + Turkey				Yes	Yes
9	Toast + Egg + Jam		Yes		Yes	
10	Waffle + Banana + Protein Shake				Yes	

If you're vegan: Template 4. Templates 5 and 6 also work if you swap whey protein powder for plant protein powder.

If you're dairy-free: Templates 4 and 9. Templates 6 and 8 work if you use plant protein powder and skip cream cheese, respectively.

If you're gluten-free: Templates 1 and (with swaps) 3 and 4 using GF pretzels.

If you're nut-free: All templates except #7 (peanut butter).

How Templates Scale: Athlete Examples

The following examples show exactly what different athletes would be served based on their body weight, workout type, and dietary needs. Portions are calculated by the solver to hit personalized carb and protein targets.

Example 1: Light Runner, Short Easy Run

Sarah – 54 kg, 45-min easy run, no restrictions

Targets: ~54 g carbs, 20 g protein

Template 1: Chocolate Milk + Banana + Protein Powder

Food	Servings	Carbs	Protein
Chocolate Milk	1 cup	30 g	9 g
Banana	1 medium	27 g	1.3 g
Protein Powder	1 scoop	3 g	24 g
Total		60 g	34.3 g

Light and quick. One cup of chocolate milk, one banana, one scoop of protein stirred in. Done in 60 seconds.

Example 2: Large Cyclist, Long Hard Ride

Marcus – 91 kg, 4-hr hard ride, no restrictions

Targets: ~109 g carbs, 36 g protein

Template 8: Bagel + Cream Cheese + Turkey

Food	Servings	Carbs	Protein
Bagel (large)	2	106 g	22 g
Cream Cheese	2 tbsp	1.6 g	1.8 g
Turkey (deli)	6 oz (2 servings)	4 g	30 g
Total		111.6 g	53.8 g

Two full bagel sandwiches. Everything scales together – more bagel means proportionally more cream cheese and turkey.

Example 3: Vegan Triathlete, Moderate Brick Workout

Aiko – 64 kg, 2-hr bike + run brick, vegan

Targets: ~64 g carbs, 20 g protein

Template 4: Plant Protein Shake + Banana + Pretzels

Food	Servings	Carbs	Protein
Plant Protein Shake	1 bottle	10 g	20 g
Banana	1 medium	27 g	1.3 g
Pretzels	1 oz	23 g	2.8 g
Total		60 g	24.1 g

Shake is fixed at 1 bottle (already meets protein target at 20 g). Banana and pretzels fill the carb gap. For a harder/longer session, add a second banana or more pretzels.

Example 4: Female Swimmer, Morning Practice, Dairy-Free

Priya – 58 kg, 1.5-hr moderate swim, dairy-free + gluten-free

Targets: ~58 g carbs, 20 g protein

Template 9: Toast + Egg + Jam (*using GF bread*)

Food	Servings	Carbs	Protein
GF Toast	2 slices	25.4 g	3.6 g
Egg (scrambled)	2 large	2 g	12.2 g
Jam	2 tbsp	27.6 g	0.2 g
Total		55 g	16 g

Protein is slightly under target at 16 g. The solver would add a small protein supplement (e.g., 1/2 scoop plant protein powder stirred into juice) to close the gap to 20 g.

Example 5: Heavy Triathlete, Ironman-Distance Race

James – 86 kg, 10+ hr Ironman, fasted start, no restrictions

Targets: ~124 g carbs, 40 g protein (max)

Template 3: Recovery Protein Drink + Pretzels

Food	Servings	Carbs	Protein
Recovery Protein Drink	1 bottle	9 g	42 g
Pretzels	4 oz (max)	92 g	11.2 g
Total		101 g	53.2 g

After an Ironman, pretzels scale to max (4 oz). One recovery drink covers protein. Carbs are still ~23 g short of the 124 g target – the solver would add a banana or additional carb source. Note: after a 10+ hr race, this is just the immediate 0-30 min recovery. A full meal follows within 1-2 hours.

Example 6: Recreational Runner, Nut Allergy

Tom – 73 kg, 1-hr moderate run, peanut/tree nut allergy

Targets: ~73 g carbs, 22 g protein

Template 10: Waffle + Banana + Protein Shake

Food	Servings	Carbs	Protein
Toaster Waffle	2 waffles	30.8 g	4 g
Banana	1 medium	27 g	1.3 g
Protein Shake (RTD)	1 bottle	5 g	25 g
Total		62.8 g	30.3 g

All templates except #7 (PB&J) are nut-free. The shake is fixed, waffles and banana scale. Close to carb target; the solver could add a tbsp of honey on the waffles for an extra 17 g carbs.

Example 7: Vegan Cyclist, Long Endurance Ride

Maya – 68 kg, 3-hr endurance ride, vegan

Targets: ~82 g carbs, 27 g protein

Template 6: Oatmeal + Protein Powder + Banana + Honey (*with plant protein powder*)

Food	Servings	Carbs	Protein
Oatmeal	1.5 servings	40.5 g	7.5 g
Plant Protein Powder	1 scoop	5 g	22 g
Banana	1 medium	27 g	1.3 g
Honey	1 tbsp	17.3 g	0.1 g
Total		89.8 g	30.9 g

Oatmeal scales up (1.5 servings for the bigger carb target). Plant protein powder swapped for whey. Warm, comforting, substantial.

Example 8: Petite Triathlete, Sprint Distance Race

Elena – 50 kg, 1.25-hr sprint triathlon, no restrictions

Targets: ~50 g carbs, 20 g protein

Template 5: Yogurt + Granola + Fruit + Honey + Protein Powder

Food	Servings	Carbs	Protein
Yogurt	1/2 cup	8.7 g	4.3 g
Granola	1/4 cup	17.5 g	3 g
Banana	1/2 medium	13.5 g	0.7 g
Honey	1/2 tbsp	8.7 g	0 g
Protein Powder	1 scoop	3 g	24 g
Total		51.4 g	32 g

Everything scales down for a lighter athlete with a shorter race. A small parfait, not a full bowl. Protein powder is still 1 full scoop (20 g protein minimum regardless of body size).

Scaling Rules Summary

Fixed vs. Scaling Components

Component Type	Behavior	Examples
RTD Protein Drinks	Fixed at 1 bottle for most scenarios. Max 1 bottle for protein_shake and plant_protein_shake. Recovery protein drink allows up to 3 for extreme cases.	Protein Shake, Plant Protein Shake, Recovery Protein Drink
Protein Powder	Fixed at 1 scoop for most athletes. 2 scoops only for athletes 90+ kg after sessions over 2.5 hr.	Protein Powder (whey or plant)
Carb Foods	Scale to hit carb target. More food for bigger athletes and longer workouts.	Banana, graham crackers, pretzels, oatmeal, toast, bagel, waffle, granola, jam, honey
Combo Foods	Scale with the carb target.	Chocolate milk (carbs + protein), yogurt (carbs + protein), milk (carbs + protein)

Linked vs. Independent Scaling

Template	Scaling Mode	What This Means
1. Choc Milk + Banana + Protein	Linked carbs, fixed protein	Milk and banana grow together. Protein powder stays at 1 scoop.
2. Shake + Banana + Grahams	Fixed protein, linked carbs	Shake = 1 bottle always. Banana and grahams grow together.
3. Recovery Drink + Pretzels	Independent	Drink = 1 bottle. Pretzels scale freely to hit carbs.
4. Plant Shake + Banana + Pretzels	Fixed protein, independent carbs	Shake = 1 bottle. Banana and pretzels each scale as needed.
5. Yogurt Parfait	Linked carbs, fixed protein	Yogurt, granola, fruit, honey grow as a unit. Protein powder = 1 scoop.
6. Oatmeal + Protein + Banana	Linked carbs, fixed protein	Oatmeal, banana, honey grow as a unit. Protein powder = 1 scoop.
7. PB&J + Milk	All linked	Everything scales together. Bigger sandwich = more of everything.
8. Bagel + CC + Turkey	All linked	Everything scales together. More bagel = more toppings.
9. Toast + Egg + Jam	All linked	Everything scales together. More toast = more eggs = more jam.
10. Waffle + Banana + Shake	Fixed protein, linked carbs	Shake = 1 bottle. Waffles and banana grow together.

Maximum Servings Per Food

Food	Max After	Notes
Protein Shake (RTD)	1 bottle	Hard cap – single bottle serving
Plant Protein Shake (RTD)	1 bottle	Hard cap – single bottle serving
Recovery Protein Drink	3 bottles	Rarely exceeds 1; up to 2 after ultra-distance
Protein Powder	2 scoops	1 scoop standard; 2 only for heavy athletes after long sessions
Chocolate Milk	3 cups	Scales from 1 to 3 cups based on carb target
Milk	3 cups	Scales from 1 to 3 cups
Yogurt	2 cups	Scales from 1/2 to 2 cups
Banana	2	1 standard; 2 for high carb targets
Graham Crackers	2 servings (4 sheets)	Scales from 2 to 4 sheets
Pretzels	4 oz	Scales freely; high sodium per serving
Oatmeal	2 servings	1 standard; 1.5-2 for high carb targets
Toast / Bread	2 slices	1-2 slices
Bagel	2	1 standard; 1.5-2 for large athletes
Toaster Waffle	2 waffles	1-2 waffles
Egg	2	2-3 standard (hard cap at 2 servings)
Peanut Butter	2 tbsp	Scales with bread
Jam / Jelly	2 tbsp	Scales with bread
Honey	2 tbsp	Scales with base food
Cream Cheese	2 tbsp	Scales with bagel
Turkey (deli)	2 servings (6 oz)	Scales with bagel
Granola	2 servings (1/2 cup)	Scales with yogurt

References

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